SkillNavora

STAGE 6 HANDBOOK

Stage 6: Enterprise projects

Outcome: Demonstrate end-to-end project ownership. Topics: Real estate: AI lead response and property matching platform, Education: AI learning assistant and student support platform, Financial services: document processing and application review automation.

From Non-Coder to Enterprise AI Builder: Learn, Build, Deploy, Get Hired and Launch Your AI Business.

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Financial figures are hypothetical examples, not advice.

Stage 6: Enterprise projects

Outcome: Demonstrate end-to-end project ownership

This handbook covers each topic of Stage 6 with a definition, why it matters, how it works, step-by-step instructions, example code or templates, common mistakes, interview questions with answers, a hands-on exercise and official resources.

Topics in this stage

1. Real estate: AI lead response and property matching platform
2. Education: AI learning assistant and student support platform
3. Financial services: document processing and application review automation

STAGE 6 · TOPIC 1

Real estate: AI lead response and property matching platform

DEFINITION

> A system for real-estate agencies that captures leads from the website, property portals and WhatsApp, qualifies them in under a minute, recommends matching properties, books site visits, follows up automatically and keeps the CRM and an agent dashboard up to date.

WHY IT MATTERS

> Agencies lose buyers when replies take hours. Fast, consistent response and follow-up raise conversion and save agents' time. It is a common first project for AI agencies and shows full FDE ownership: discovery, integration, AI, deployment and adoption.

HOW IT WORKS

- Channels: website form (webhook), portal lead emails (email parsing), WhatsApp Business Platform (official API via Meta or a provider).
- Backend: FastAPI service with PostgreSQL (leads, properties, visits, messages, agents) and a job queue for follow-ups.
- AI: an LLM extracts intent, budget, location, BHK and timeline; drafts replies in the lead's language; a scoring step marks hot/warm/cold.
- Matching: SQL filters (budget, location, BHK) plus semantic search on property descriptions.
- Workflow: hot leads assigned to an agent instantly; visits booked through a calendar; follow-ups on day 1, 3 and 7; CRM updated.
- Dashboard: React app for agents (lead list, conversation, suggested reply) and managers (response time, conversion, source performance).
- Controls: human approval for first messages until quality is proven, WhatsApp template and opt-in rules, consent and data retention, audit log.

STEP-BY-STEP

1. Discovery: map the current lead journey, sources, volumes, response times and CRM.
2. Data model: leads, properties, agents, visits, messages, events.
3. Build intake webhooks and the extraction prompt with structured output; evaluate on 100 past enquiries.

4. Build matching and reply drafting; add approval queue in the dashboard.
5. Integrate the calendar, CRM and WhatsApp; add follow-up jobs.
6. Deploy (Docker + Cloud Run/App Runner/Container Apps), add monitoring, train agents, and measure response time and conversion for 4 weeks.

EXAMPLE CODE / TEMPLATE

```
# Lead extraction with structured output (Claude) - core of the intake step
import anthropic, json
from pydantic import BaseModel

client = anthropic.Anthropic()

class LeadInfo(BaseModel):
    intent: str            # buy | rent | sell | other
    budget_inr: int | None
    locations: list[str]
    bhk: int | None
    timeline: str | None   # e.g. 'this month', '3-6 months'
    language: str          # reply in this language

SYSTEM = ('Extract real-estate enquiry details. Use null when a value is not stated. '
          'Convert lakh/crore to rupees (1 lakh = 100000, 1 crore = 10000000).')

def extract(message: str) -> LeadInfo:
    msg = client.messages.parse(
        model='claude-opus-5-5', max_tokens=1000, system=SYSTEM,
        messages=[{'role': 'user', 'content': message}],
        output_format=LeadInfo,
    )
    return msg.parsed_output

def score(info: LeadInfo) -> str:
    if info.intent in ('buy', 'rent') and info.budget_inr and info.timeline in ('this month',
'immediately'):
        return 'hot'
    return 'warm' if info.intent in ('buy', 'rent') else 'cold'

# Matching (SQL): SELECT * FROM properties WHERE price BETWEEN :b*0.85 AND :b*1.1
#                 AND area = ANY(:locations) AND (:bhk IS NULL OR bhk = :bhk) LIMIT 3;
```

COMMON MISTAKES

- Sending WhatsApp messages outside the platform's policy (no opt-in, no approved template).
- Auto-sending AI replies before quality is proven.
- Recommending properties that are no longer available (stale inventory).
- Not measuring the baseline response time and conversion, so impact cannot be shown.

INTERVIEW QUESTIONS AND ANSWERS

Q. How would you demonstrate business impact?

A. Measure baseline median response time, contact rate, visit rate and conversion for 4 weeks, then the same after launch, with the same sources and seasons noted; report time saved per agent and revenue influenced, labelled as estimates.

Q. What are the main risks and controls?

A. Wrong or invented property details (ground answers in inventory data), privacy and messaging rules (consent, templates, retention), over-automation (approval queue, easy hand-off to an agent), and integration failures (retries, monitoring).

HANDS-ON EXERCISE

> Beginner: website form → LLM extraction → email to agent with a suggested reply. Intermediate: add property matching, CRM sync and a React dashboard with an approval queue. Advanced: add WhatsApp, multilingual replies, scheduled follow-ups, visit booking, analytics, evaluation in CI and multi-tenant SaaS for several agencies.

OFFICIAL AND RECOMMENDED RESOURCES

- WhatsApp Business Platform docs https://developers.facebook.com/docs/whatsapp/
- Claude: structured outputs https://docs.claude.com/en/docs/build-with-claude/structured-outputs
- FastAPI tutorial https://fastapi.tiangolo.com/tutorial/
- Google Calendar API https://developers.google.com/workspace/calendar/api/guides/overview

Education: AI learning assistant and student support platform

DEFINITION

A platform for schools, colleges or edtech companies that answers student questions from course material with citations, explains concepts at the student's level, tracks progress, recommends next lessons, supports admissions and fee queries, and gives teachers dashboards of where students struggle.

WHY IT MATTERS

Teachers cannot answer every question at every hour, and admissions teams repeat the same answers. A grounded assistant gives 24/7 help, while learning analytics show teachers where to intervene. It demonstrates RAG, personalisation, analytics and responsible AI with minors' data.

HOW IT WORKS

- Content ingestion: syllabus, notes, slides and past papers parsed, chunked and indexed with course, chapter and level metadata.
- Student assistant: RAG over the student's own courses only; answers cite the lesson; explains step by step; encourages attempts before giving full solutions (configurable by the teacher).
- Progress tracking: quiz attempts, topics viewed and questions asked stored per student.
- Recommendations: next lesson or practice set based on weak topics.
- Teacher dashboard: common confusions, struggling students, question themes.
- Admissions and fees: an FAQ assistant over official policy documents, reminders through approved channels, and hand-off to a counsellor.
- Controls: consent from parents/guardians for minors, data minimisation, no use of student data for model training, content safety filters, teacher review of AI-generated quizzes, counselling decisions remain with humans.

STEP-BY-STEP

1. Discovery with teachers, students and admin staff: top questions, materials, devices, languages.
2. Ingest one course; build RAG with citations; evaluate on 100 real student questions graded by teachers.
3. Add student accounts, course enrolment and permission-filtered retrieval.
4. Add quizzes, progress records and topic-mastery estimates.
5. Build the teacher dashboard and the admissions/fees FAQ assistant.
6. Pilot with one class for 4 weeks; measure usage, answer accuracy, teacher time saved and student satisfaction.

EXAMPLE CODE / TEMPLATE

```
# Tutor-style answer grounded in course material, adapted to level
import anthropic
client = anthropic.Anthropic()

TUTOR_RULES = '''You are a patient tutor for {course} ({level}).
- Use only the lesson extracts provided; cite them like [Lesson 3.2].
- Explain step by step in simple language; use one everyday example.
- If the student asks for a homework answer, give a hint and the first step, then ask them to try.
```

```
- If the extracts do not cover the question, say so and suggest asking the teacher.'''

def tutor_answer(question: str, extracts: list[dict], course: str, level: str) -> str:
    ctx = '\n'.join(f"<extract lesson='{e['lesson']}'>{e['text']}</extract>" for e in extracts)
    msg = client.messages.create(
        model='claude-opus-5-5', max_tokens=1500,
        system=TUTOR_RULES.format(course=course, level=level),
        messages=[{'role': 'user', 'content': f'<extracts>{ctx}</extracts>\n<question>{question}
</question>'}])
    return ''.join(b.text for b in msg.content if b.type == 'text')

# Mastery update after a quiz (simple moving average per topic)
def update_mastery(old: float, correct: bool, weight: float = 0.3) -> float:
    return round((1 - weight) * old + weight * (1.0 if correct else 0.0), 2)
```

COMMON MISTAKES

- Answering from general knowledge instead of the school's syllabus.
- Giving full homework answers, which teachers reject.
- Collecting more student data than needed or without guardian consent.
- No teacher involvement in evaluation and rollout.

INTERVIEW QUESTIONS AND ANSWERS

Q. How do you keep an education assistant safe and appropriate?

A. Ground answers in approved material, filter unsafe content, restrict topics, involve teachers in rules and review, protect minors' data with consent and minimisation, log interactions for review, and escalate wellbeing concerns to humans.

Q. How would you measure learning impact?

A. Compare quiz and exam performance, homework completion and engagement for pilot and comparison groups, plus teacher time saved and student feedback, being careful about other factors.

HANDS-ON EXERCISE

> Beginner: course-material Q&A with citations. Intermediate: student accounts, quizzes, progress tracking and teacher dashboard. Advanced: personalised learning paths, multilingual explanations, admissions/fees assistant with WhatsApp reminders, LMS integration and evaluation pipeline.

OFFICIAL AND RECOMMENDED RESOURCES

- UNESCO: guidance for generative AI in education https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research
- Anthropic: contextual retrieval https://www.anthropic.com/news/contextual-retrieval
- Moodle developer docs (LMS integration) https://moodledev.io/
- India DPDP Act (official text) https://www.meity.gov.in/data-protection-framework

Financial services: document processing and application review automation

DEFINITION

A system for lenders, insurers or fintechs that ingests application documents (ID, bank statements, salary slips, invoices, claims forms), extracts and validates key fields, checks completeness and consistency against rules, flags exceptions, prepares a review summary, and routes every case to authorised human reviewers with a full audit trail.

WHY IT MATTERS

Manual document review is slow, costly and error-prone. Automating extraction and checks shortens turnaround and lets reviewers focus on judgement. Because decisions affect people's finances and are regulated, the design must keep consequential decisions with authorised humans and be fully auditable.

HOW IT WORKS

- Ingestion: secure upload portal or API; files stored encrypted in object storage with retention rules.
- Classification: identify document type (PAN/ID, bank statement, salary slip, ITR, invoice, claim form).
- Extraction: OCR where needed, then an LLM with a strict schema extracts fields with page references and confidence.
- Validation: deterministic rules in code (formats, date ranges, totals add up, name matches across documents, income consistency).
- Exception flags: missing documents, inconsistencies, unusual patterns; never automatic rejection.
- Review workspace: reviewer sees extracted fields beside the source page, the flags and an AI summary; edits are tracked; the reviewer records the decision.
- Governance: role-based access, PII masking in logs, data residency, model and prompt versioning, audit trail of every extraction and human decision, regulatory requirements (for example RBI guidelines in India).

STEP-BY-STEP

1. Discovery with operations, risk, compliance and IT: document types, volumes, rules, systems, regulations.
2. Collect a labelled sample (with permission and masking) of 200 document sets.
3. Build classification and extraction with schemas; measure field-level accuracy.
4. Implement validation rules in code with the risk team; build exception flags.
5. Build the reviewer workspace with side-by-side source view and audit logging.
6. Security review, UAT with reviewers, phased rollout; monitor accuracy, turnaround time and reviewer overrides.

EXAMPLE CODE / TEMPLATE

```
# Extract fields from a salary slip, then validate with deterministic rules
import anthropic, base64
from pydantic import BaseModel

client = anthropic.Anthropic()
```

```
class SalarySlip(BaseModel):
    employee_name: str
    employer_name: str
    month: str               # YYYY-MM
    gross_pay: float
    net_pay: float
    deductions_total: float
    page: int                # page where values were found

def extract_slip(pdf_path: str) → SalarySlip:
    data = base64.standard_b64encode(open(pdf_path, 'rb').read()).decode()
    msg = client.messages.parse(
        model='claude-opus-5-5', max_tokens=2000,
        system='Extract values exactly as printed. Do not infer missing values.',
        messages=[{'role': 'user', 'content': [
            {'type': 'document', 'source': {'type': 'base64', 'media_type': 'application/pdf', 'data':
data}},
            {'type': 'text', 'text': 'Extract the salary slip fields.'}]}],
        output_format=SalarySlip,
    )
    return msg.parsed_output

def validate(slip: SalarySlip, applicant_name: str) → list[str]:
    flags = []
    if abs(slip.gross_pay - slip.deductions_total - slip.net_pay) > 1:
        flags.append('Net pay does not equal gross minus deductions')
    if applicant_name.lower() not in slip.employee_name.lower():
        flags.append('Name on slip does not match application')
    return flags   # flags go to a human reviewer; the system never rejects on its own
```

COMMON MISTAKES

- Letting the model make approve/reject decisions.
- Using the model for arithmetic checks instead of code.
- Storing documents or logs with unmasked personal and financial data.
- No field-level accuracy measurement before rollout.

INTERVIEW QUESTIONS AND ANSWERS

Q. Why keep humans in the decision?

A. Credit and claims decisions are consequential and regulated; models can misread documents or be manipulated. Automation should speed up extraction and checks, while authorised reviewers make and are accountable for decisions.

Q. How do you measure extraction quality?

A. Field-level precision and recall against a labelled set, per document type, plus reviewer correction rates in production; set thresholds per field (stricter for amounts and identifiers).

HANDS-ON EXERCISE

> Beginner: extract fields from one document type with a schema and show them next to the PDF. Intermediate: multiple document types, cross-document validation rules and a reviewer queue. Advanced: full case workflow with audit trail, role-based access, PII masking, evaluation pipeline, monitoring of reviewer overrides and deployment in the customer's cloud.

OFFICIAL AND RECOMMENDED RESOURCES

- Claude: PDF support https://docs.claude.com/en/docs/build-with-claude/pdf-support
- RBI: master directions (official) https://www.rbi.org.in/Scripts/BS_ViewMasterDirections.aspx

- NIST AI Risk Management Framework https://www.nist.gov/itl/ai-risk-management-framework
- OWASP Top 10 for LLM Applications https://genai.owasp.org/llm-top-10/